



TOE-YOLO: accurate and efficient detection of tiny objects in UAV imagery

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Abstract

Existing models achieve high accuracy in small object detection but often overlook the rotational and dense characteristics of targets. To address this, we propose an improved YOLO11-based model specifically designed for detecting dense and rotated small objects in UAV scenarios. To better extract features from rotated targets, we design a C3k2-ARC module that enhances the model's rotational detection capability. In addition, we introduce the CL-Concat feature fusion module, which combines traditional concatenation with channel and spatial attention, significantly improving the quality of multi-scale feature fusion. The experimental results demonstrate that the proposed method achieves notable performance improvements across multiple public benchmark data sets. Compared with the advanced YOLO11n model, our approach achieves gains of 1.6% on VISDRONE, 1.4% on UAVDT, 0.2% on CARPK, and 0.1% on UAVROD, further validating its effectiveness across diverse UAV detection scenarios.

Keywords Tiny object detection · High-altitude UAV imagery · Context-aware feature fusion · Rotated object detection

1 Introduction

In recent years, the advancement of deep learning has significantly accelerated the industrial adoption of unmanned aerial vehicles, enabling their deployment across a wide range of applications [1], including traffic surveillance [2], precision agriculture [3], logistics [4], and disaster management [5]. Among various deep learning technologies, real-time object detection stands out as one of the most extensively utilized in UAV-based systems. With the emergence of edge computing and cloud-based processing frameworks [6],

the computational burden has gradually shifted away from onboard processing. This transition has reduced the reliance on local hardware resources and made it feasible to deploy object detection models with moderate resource demands directly on UAV platforms. Consequently, the development of object detection algorithms that strike a balance between high accuracy and computational efficiency has become a critical requirement for UAV-based applications.

However, when directly applying existing models to object detection tasks involving UAV-captured imagery, several critical challenges typically arise, as clearly illustrated in the examples shown in Fig. 1. First, UAV images often cover wide areas with small and indistinct objects, which increases the likelihood of false detections. Second, these scenes frequently exhibit high object density, resulting in significant occlusion. Third, due to the overhead perspective of UAVs, many detected objects exhibit strong orientation characteristics, often appearing in rotated forms that standard detection models struggle to handle effectively. Fourth, although some models demonstrate promising detection performance on aerial imagery, their large model sizes or slow inference speeds make them impractical for deployment on UAV platforms. Collectively, these four issues present major challenges for achieving accurate and efficient object detection in UAV-based applications.

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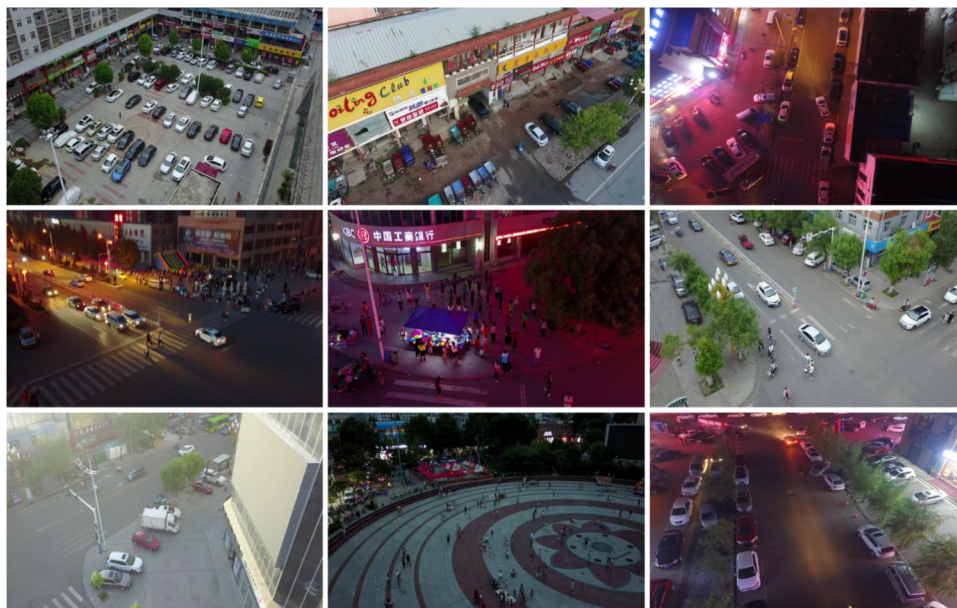
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Fig. 1 Visualization of UAV-captured images



To address the numerous challenges of UAV-based object detection, many researchers have proposed various YOLO-based improvements tailored specifically for aerial imagery. Luo et al. [7] introduced YOLO-UAV, which enhances detection performance by incorporating an Efficient Channel Attention module after the Focus layer and designing three feature extraction modules based on asymmetric convolutions. Although this method achieved improvements over its baseline model in certain metrics, it is built upon YOLOv5l, a relatively large and outdated architecture—resulting in a noticeable gap in overall performance compared to state-of-the-art models, and making it difficult to deploy on resource-constrained UAV platforms. Wang et al. [8] proposed SMFF-YOLO, which utilizes a Bidirectional Feature Fusion Pyramid (BFFP) and an Adaptive Atrous Spatial Pyramid Pooling (AASPP) module to improve multi-scale object detection performance. While this model achieves remarkable accuracy improvements, the AASPP module introduces a substantial number of parameters and high computational complexity (GFLOPs), leading to a significant increase in model size and hindering the balance between accuracy and model efficiency. Min et al. [9] presented YOLO-DCTI, which enhances detection of small and tiny objects by designing a decoupled contextual transformer-based detection head (DCTI). Although it delivers strong performance in UAV-based small object detection, the model mainly focuses on tiny objects and does not sufficiently address the presence of multi-scale objects commonly found in high-altitude imagery, thereby limiting its general applicability. These studies underscore the promising potential of YOLO-series models for UAV-based object detection. However, most of the existing improvements struggle to achieve an effective

balance among model size, inference speed, and detection accuracy. Furthermore, they often overlook a critical characteristic of high-altitude UAV imagery: the widespread presence of rotated objects across multiple scales. This oversight may significantly limit the practical applicability and robustness of such models in real-world UAV detection tasks.

Inspired by previous research, this paper proposes an efficient neural network model named TOE-YOLO, based on the lightweight YOLOv11n architecture, aiming to address key challenges in high-altitude image object detection. First, considering that multi-scale objects in aerial imagery often exhibit rotational characteristics, we design a novel C3k2-ARC module, built upon the Adaptive Rotated Convolution (ARC) [10], and integrate it into the backbone of YOLOv11n. This integration enhances the model's ability to extract features from rotated objects while slightly reducing both the parameter count and computational complexity. Furthermore, to alleviate the issues of feature redundancy and information conflict that may arise during feature fusion in the concatenation operations within the YOLO neck structure, especially for multi-scale targets, we introduce an attention-integrated concatenation module named CL-Concat, inspired by the CPAM module [11]. In CL-Concat, a channel attention mechanism is applied to each feature map prior to concatenation, allowing the network to emphasize informative channels and suppress irrelevant ones. This is followed by a position attention mechanism after concatenation, enabling the model to focus on spatially critical regions. Together, these strategies improve the quality of feature fusion and significantly enhance the detection performance for small and densely distributed objects.

Our main contributions are listed as follows:

- To address the characteristics of multi-scale rotated objects in aerial images, this paper proposes the novel C3k2-ARC module. This module significantly enhances the YOLO backbone's capability to extract features from multi-scale rotated objects while achieving model size reduction without compromising performance.
- Considering that the commonly used concatenation operation in the neck may cause feature redundancy and information conflicts during multi-scale feature fusion, we introduce the CL-Concat module. By employing channel and spatial attention mechanisms, CL-Concat enhances the expression of critical information and facilitates effective integration of multi-scale features, thereby improving the detection accuracy of multi-scale targets.
- We validate our approach on multiple data sets. Experimental results demonstrate that although the TOE-YOLO model reduces both parameter count and GFLOPs, it achieves significant improvements across various metrics compared to the baseline model and several state-of-the-art methods.

2 Related work

2.1 Traditional object detection

Deep learning-based object detection techniques are generally divided into two major paradigms: two-stage and one-stage detection frameworks.

Two-stage detectors operate by first generating a set of candidate regions that potentially contain objects. These proposals FCNare then subjected to further classification and bounding box regression to refine both the object category and its precise location. Representative models such as Faster R-CNN [12] and R-FCN [13] fall into this category. These methods are well-regarded for their high detection accuracy and robust localization capabilities. However, due to their multi-step processing pipeline, they often incur significant computational costs and exhibit slower inference speeds, making them less ideal for time-sensitive or real-time applications.

In contrast, one-stage detectors like YOLO (You Only Look Once) [14] bypass the proposal generation phase and directly predict class probabilities and bounding box coordinates in a single pass through the network. This unified approach greatly enhances inference efficiency, making it highly suitable for deployment in resource-constrained environments and applications, where rapid response is critical. As a result, one-stage methods have become increasingly popular in real-time scenarios, such as aerial object detection using drones.

2.2 Rotated object detection

Rotated object detection has emerged as a critical task within the broader field of object detection, owing to the prevalence of objects exhibiting significant orientation variations in real-world scenarios. These variations pose substantial challenges for conventional detection frameworks, particularly in the accurate extraction and representation of rotation-sensitive features.

Existing research efforts have primarily approached this challenge from three key directions. The first line of work focuses on the development of dedicated detectors tailored for rotated objects. For example, Xie et al. [15] proposed the Oriented Region Proposal Network (RPN), which explicitly incorporates rotation into the proposal generation stage to improve detection accuracy. The second direction involves enhancing object representation schemes. Xu et al. [16], for instance, extended the conventional horizontal bounding box paradigm by introducing four gliding offset parameters, enabling a more flexible and precise modeling of arbitrarily oriented objects. The third line of research centers on the design of loss functions suitable for rotated object regression. Yang et al. [17] reformulated rotated bounding boxes as two-dimensional Gaussian distributions and employed distance metrics such as Gaussian Wasserstein distance and Kullback–Leibler divergence to improve regression accuracy.

While these approaches have contributed significantly to the advancement of rotation-aware detectors, they often underexplore the role of the backbone feature extractor, which is crucial for robust representation learning. Although rotation-equivariant operations can retain orientation information, they often fall short in capturing intra-image and cross-domain orientation variations, thereby limiting both detection accuracy and model generalization.

2.3 Object detection in aerial images

Aerial imagery presents substantial challenges for computer vision tasks due to variations in acquisition conditions, such as differences in time and location of capture. These variations often lead to issues, including poor or excessive lighting, the presence of multi-scale and small objects, densely packed targets, and arbitrary object orientations. Such complexities make the analysis of aerial images both a critical and technically demanding endeavor in the field of visual perception.

To improve detection performance in aerial scenarios, Zhu et al. [18] proposed TPH-YOLOv5, an enhanced variant of YOLOv5 that incorporates Transformer Prediction Heads (TPH) to better capture complex spatial relationships and improve object detection performance in UAV-based applications. Similarly, Mu et al. [19] introduced CSC-YOLO,

which enhances small object detection by integrating an additional detection head, a cross-channel fusion module, and a multi-level feature fusion downsampling structure.

While these approaches have effectively addressed the limitations of high-altitude object detection from various perspectives, they often incur substantial computational costs and primarily focus on small object detection. To address these limitations, this study aims to develop a light-weight yet accurate model capable of robust multi-scale object detection, thereby ensuring computational efficiency and practical deployability on resource-constrained UAV platforms.

3 Proposed method

This section provides a comprehensive overview of the proposed TOE-YOLO architecture, detailing its overall design as well as the enhanced modules C3k2-ARC and Channel and Location-aware Concatenation (CL-Concat).

3.1 Overall structure of TOE-YOLO

Although YOLOv11 has demonstrated strong performance across various general-purpose data sets, its smallest variant, YOLOv11n still faces notable limitations when applied to multi-scale object detection and complex visual environments. These challenges become particularly pronounced in drone-based aerial imagery, where objects often appear at varying scales and exhibit significant orientation, resulting in numerous rotated targets. Moreover, due to the high-altitude vantage point of UAV platforms, the captured images frequently contain densely distributed small objects, further complicating the detection task. To effectively address these issues, we propose an enhanced object detection framework, termed TOE-YOLO, specifically designed to improve robustness and accuracy under the constraints typical of aerial imaging scenarios.

Figure 2 illustrates the architecture of the proposed TOE-YOLO network. Built upon the lightweight YOLOv11n framework, this enhanced neural network integrates the C3k2-ARC module into its backbone, which significantly strengthens the model's ability to extract discriminative features from rotated objects commonly found in aerial imagery. This design not only improves detection performance but also contributes to a reduction in overall model size and an increase in inference speed.

In addition, to address potential issues such as feature redundancy and semantic conflict that may arise during the fusion of multi-scale features, we propose the Channel and Location-aware Concatenation (CL-Concat) module as a replacement for the conventional concatenation operation in the original YOLO neck. This module introduces a

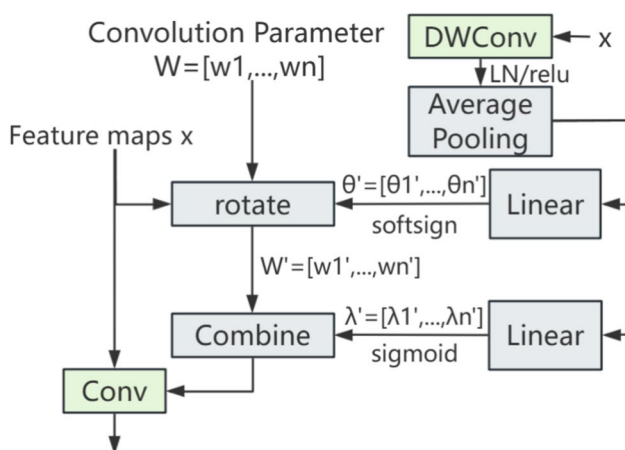
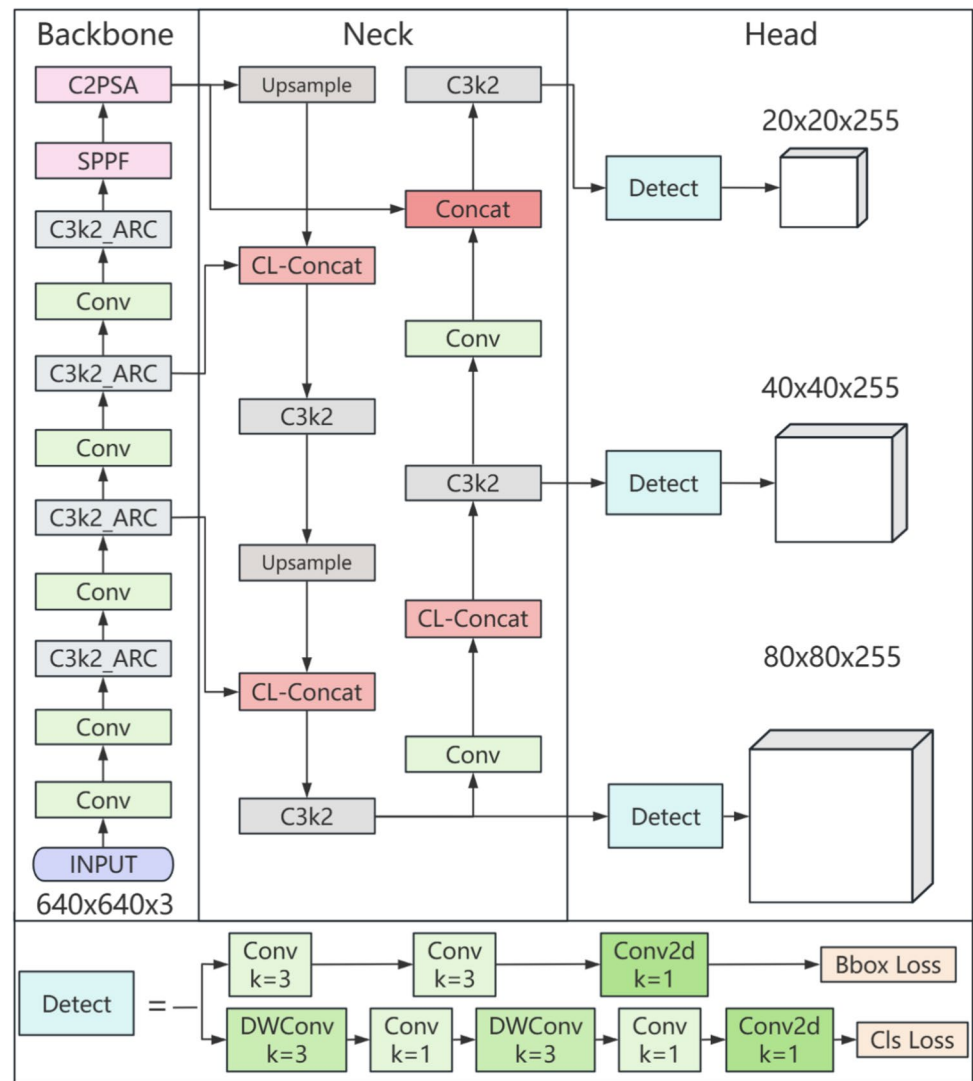
channel attention mechanism prior to feature fusion to filter out redundant or uninformative channels, followed by a position attention mechanism after fusion to guide the network's focus toward spatially significant regions. Through this two-stage attention refinement strategy, CL-Concat effectively enhances the quality of feature integration and improves the detection accuracy for multi-scale objects.

3.2 C3k2-ARC

A fundamental challenge in rotated object detection lies in the significant variation of object orientations across different images, and even within a single image, where multiple directional instances may coexist. Traditional approaches often attempt to address this issue by employing larger or more complex convolutional kernels to enhance rotational adaptability. However, such strategies typically result in increased computational burden and decreased inference efficiency, limiting their practicality in real-time or resource-constrained applications. Another common approach is to employ rotated bounding boxes to achieve high-precision detection of rotated targets. This method allows for more accurate description and localization of objects in arbitrary orientations while reducing background interference. However, rotated bounding box techniques typically involve complex proposal generation and regression processes, resulting in significantly increased computational overhead. Moreover, to maintain high detection accuracy, such models often require more parameters and higher memory bandwidth during both training and inference, making them difficult to deploy efficiently on resource-constrained UAV platforms.

To overcome these limitations, Pu et al. [10] introduced the Adaptive Rotated Convolution (ARC) module, which enables convolutional kernels to dynamically rotate based on the orientation of the input features. This design eliminates the need for large kernel sizes and effectively reduces computational complexity while enhancing the model's ability to capture rotation-sensitive features.

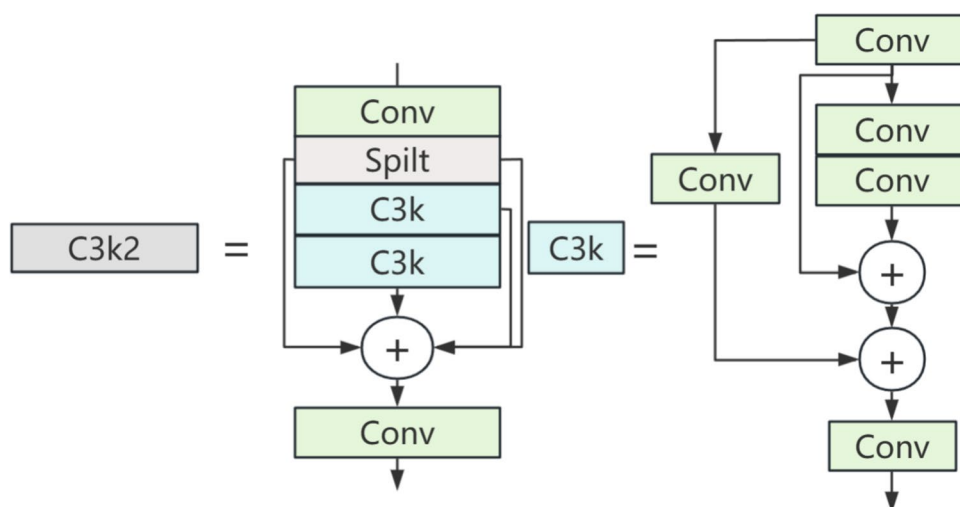
As illustrated in Fig. 3, the ARConv module incorporates kernel rotation, feature composition, and linear mapping operations to achieve adaptive orientation alignment. Specifically, it adjusts the position and parameters of the convolutional kernels according to the directional characteristics of the input, allowing the network to flexibly respond to objects with diverse orientations. This dynamic mechanism significantly improves the detection accuracy of rotated objects, which traditional convolutional operations often struggle to handle. Moreover, by aggregating convolutional responses across multiple angles through learnable weighted fusion, the ARConv module further strengthens the network's capacity to perceive and represent objects from various directions, thereby enhancing its robustness in complex rotational scenarios.

Fig. 2 Structure diagram of TOE-YOLO**Fig. 3** Structure diagram of ARConv module

As illustrated in Fig. 4, the original C3k2 module primarily consists of a series of standard convolutional layers, C3k blocks, and feature concatenation operations. The C3k structure adopts a bottleneck-based design, leveraging split-and-merge strategies to efficiently capture local features. This enables the network to enhance nonlinear representation capabilities and expand the receptive field while maintaining computational efficiency. As a result, the C3k2 module has been widely adopted in lightweight detection networks due to its effectiveness in feature extraction under conventional scenarios.

However, aerial imagery captured from high altitudes presents additional challenges. Targets in such images often exhibit significant rotation and orientation variations, making it difficult for standard convolution kernels, which operate with fixed directions, to extract representative features effectively. Moreover, the C3k2 module lacks the ability to explicitly model orientation, limiting its

Fig. 4 Structure diagram of C3k2 module



robustness in scenarios, where rotated objects are densely distributed and diverse in angle.

To address these limitations, this paper introduces an improved module named C3k2-ARC, as depicted in Fig. 5. Inspired by the Adaptive Rotated Convolution (ARC), C3k2-ARC is specifically designed to improve detection performance for rotated objects in aerial scenes. In this design, an ARCBLOCK is developed and integrated into the C3k2 structure, enhancing directional feature extraction while maintaining the lightweight nature of the model.

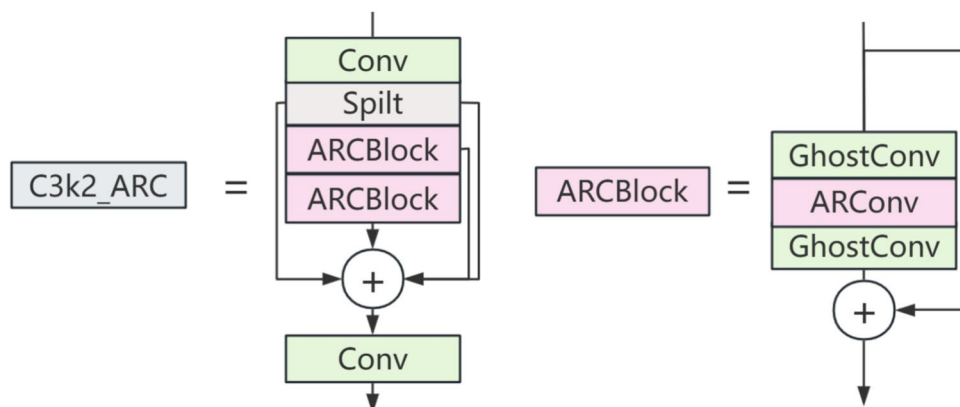
The ARCBLOCK is composed of GhostConv and ARConv layers. ARConv dynamically adjusts the convolutional kernel orientation based on the directional information of the input features, enabling effective extraction of rotationally variant target features and overcoming the limitations of conventional convolutions in perceiving multi-directional objects. GhostConv employs a lightweight mechanism to generate redundant features, significantly reducing the model's parameter count and computational complexity while maintaining expressive capacity, thereby enhancing overall inference efficiency. Furthermore, we introduce residual connections within the ARCBLOCK to facilitate efficient feature

propagation and improve training stability. This architecture not only enhances the model's ability to perceive multi-directionally rotated targets but also strengthens the feature representation of rotated objects across multiple scales and complex backgrounds, resulting in improved overall detection performance and robustness.

3.3 Channel and location-aware concatenation module (CL-Concat)

In multi-scale object detection tasks, feature maps extracted from different hierarchical levels exhibit significant disparities in both semantic abstraction and spatial resolution. Directly concatenating these heterogeneous features without appropriate alignment and guidance often leads to semantic inconsistency and the loss of fine-grained details, thereby compromising the effectiveness of feature fusion. Moreover, such inconsistency is particularly detrimental to small object detection, as their already weak discriminative cues are easily overwhelmed by high-level abstract semantics during the fusion process.

Fig. 5 Structure diagram of C3k2-ARC module



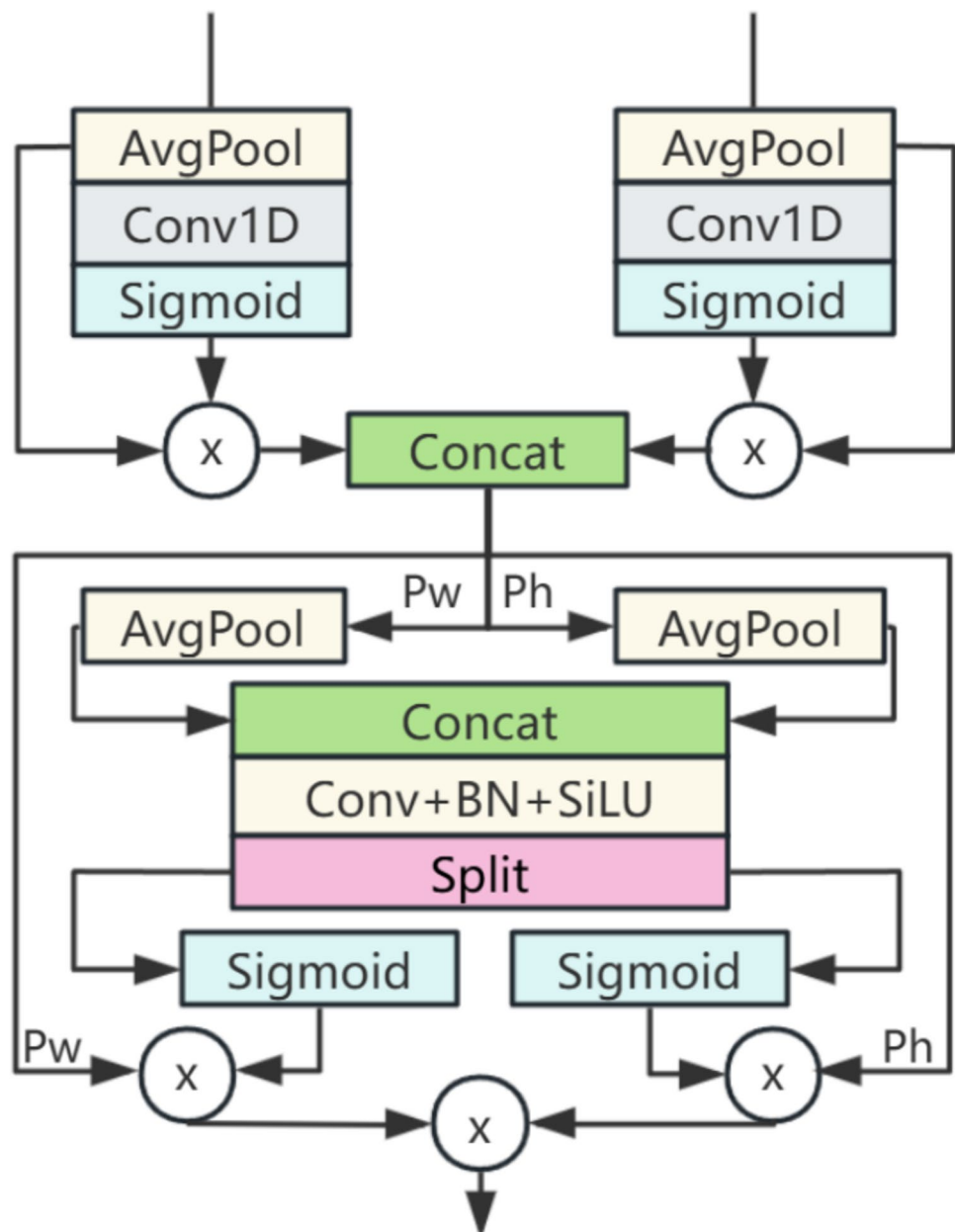
To address the above challenges, we propose an attention-integrated feature concatenation module, termed CL-Concat (Channel and Location-aware Concatenation). As illustrated in Fig. 6, the module is composed of two sequential attention mechanisms, channel attention and location (spatial) attention, which collaboratively optimize the fusion of multi-scale features.

In the channel attention stage, each input feature map is processed independently through its own attention branch. Each branch first applies global average pooling to compress the spatial dimensions and extract a channel-level semantic descriptor. This descriptor is then passed through a 1D convolution layer with an adaptively determined kernel size, which captures local dependencies across channels.

The output is activated via a sigmoid function to generate a channel attention map. This attention map is then applied to the original feature map via elementwise multiplication, enhancing informative channels while suppressing redundant ones. The refined feature maps from all branches are then concatenated along the channel dimension, forming a unified channel-enhanced representation. This mechanism preserves the representational capacity of each scale while improving the consistency and selectivity of semantic information across them.

Next, the fused features are passed into the location attention module, which adopts a lightweight coordinate-separated modeling structure. Specifically, the module performs average pooling along the horizontal and vertical directions

Fig. 6 Structure diagram of channel and location-aware concatenation module (CL-Concat)



to obtain spatial context features for each axis independently. These direction-specific features are concatenated along the channel dimension and processed through a 1×1 convolution, followed by Batch Normalization and SiLU activation, to extract direction-aware representations. The output is then split into horizontal and vertical attention maps, which are, respectively, applied to the original features via elementwise multiplication to achieve axis-specific spatial enhancement. This approach effectively captures the spatial distribution of elongated or densely packed objects while maintaining low computational and parameter overhead.

In summary, CL-Concat employs a two-stage attention strategy-channel refinement followed by spatial enhancement to improve the consistency of multi-scale semantic fusion while preserving spatially critical information. This design not only performs well in scenarios with significant scale variation, but also substantially enhances localization accuracy and detection robustness for dense and small objects. Moreover, due to the adoption of lightweight attention mechanisms and efficient feature fusion strategies in this structure, the model achieves significant performance improvements with negligible increases in parameters and computational cost.

3.4 Loss function

The bounding box regression in this work utilizes the YOLO-based CIoU loss. The computation of CIoU, as shown in the following formulas, extends the traditional IoU by incorporating the distance between the centers of the predicted and ground truth boxes, the diagonal length of the minimum enclosing box, and the aspect ratio consistency, thereby enabling more precise guidance for bounding box regression:

$$\text{IoU} = \frac{\text{Area}(\mathbf{B}_p \cap \mathbf{B}_{gt})}{\text{Area}(\mathbf{B}_p \cup \mathbf{B}_{gt})} \quad (1)$$

$$\rho(\mathbf{b}_p, \mathbf{b}_{gt}) = \sqrt{(x_p - x_{gt})^2 + (y_p - y_{gt})^2} \quad (2)$$

$$c = \sqrt{(\max(x_{\max}^p, x_{\max}^{gt}) - \min(x_{\min}^p, x_{\min}^{gt}))^2 + (\max(y_{\max}^p, y_{\max}^{gt}) - \min(y_{\min}^p, y_{\min}^{gt}))^2} \quad (3)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w_{gt}}{h_{gt}} - \arctan \frac{w_p}{h_p} \right)^2 \quad (4)$$

$$\alpha = \frac{v}{(1 - \text{IoU}) + v} \quad (5)$$

$$\text{CIoU} = \text{IoU} - \frac{\rho^2(\mathbf{b}_p, \mathbf{b}_{gt})}{c^2} - \alpha v \quad (6)$$

$$\mathcal{L}_{\text{CIoU}} = 1 - \text{CIoU} \quad (7)$$

Specifically, CIoU computes the overlap between a given predicted box $\mathbf{B}_p$ and ground truth box $\mathbf{B}_{gt}$, namely, the IoU, which measures the degree of region alignment between the prediction and the ground truth. Next, the Euclidean distance ρ between the centers of the predicted and ground truth boxes is calculated to quantify the center offset. At the same time, the diagonal length of the minimum enclosing box c is defined to normalize the contribution of the center distance to the loss, thus considering the maximal spatial discrepancy between the two boxes.

Moreover, to further constrain the aspect ratio consistency, CIoU introduces the term v to measure the difference in width–height ratios between the predicted and ground truth boxes. A weighting factor α is then employed to balance the influence of the aspect ratio based on the IoU and v . By incorporating these constraints, the CIoU loss simultaneously accounts for overlap, center offset, and aspect ratio, resulting in improved convergence and bounding box regression accuracy.

4 Experiment settings

This study utilizes four diverse and representative UAV image data sets to evaluate the effectiveness of the proposed method: VisDrone2019 [20], UAVDT [21], CARPK [22], and UAV-ROD [23]. In this section, we provide an overview of these data sets, followed by a description of the experimental configuration and evaluation criteria.

4.1 Data sets

The VisDrone2019 data set is designed for object detection tasks using aerial imagery captured by UAVs. It comprises a total of 10,209 static images, partitioned into 6471 for

training, 548 for validation, and 3190 for testing. The data set includes ten object categories: pedestrian, bus, motor, bicycle, person, car, truck, van, awning-tricycle, and tricycle. A notable challenge of this data set is the presence of small-scale targets and densely distributed objects. The typical image resolution is approximately 2000×1500 pixels.

The UAVDT data set features drone-captured imaUAVgery over urban environments under various

weather conditions and altitudes, offering a complex scenario for object detection. It includes 25,137 training images and 15,598 test images, covering three classes: car, truck, and bus. The average resolution is 1080×540 pixels.

The CARPK data set is one of the largest aerial parking lot data sets, containing around 90,000 car instances annotated across four different parking areas. Each image includes detailed bounding box annotations, with the most crowded scene featuring up to 188 vehicles. For our experiments, we utilized 989 images for training and 459 for testing. The image resolution is 1280×720 pixels.

The UAV-ROD data set focuses on vehicle detection in traffic segments and parking zones, also using UAV-captured images. Each vehicle is labeled with bounding boxes, with a maximum of 75 cars in a single frame. In our setup, 1150 images were used for training and 427 for testing. The data set offers high-resolution imagery at 2720×1530 pixels.

4.2 Network training

The training process for this study was conducted on a development environment built on Ubuntu 22.04.4 LTS. The system was powered by a tenth-generation Intel Core i9-10850K processor, capable of reaching a maximum clock speed of 5.20 GHz. An NVIDIA GeForce RTX 3090 GPU with 24 GB of VRAM was used to accelerate computations. The experiments utilized CUDA version 11.3 and were implemented using Python 3.8 in combination with PyTorch version 1.10.0.

For training, we adopted a batch size of 16 and ran the model for 200 epochs. The Adam optimizer was used with an initial learning rate set to 0.001. To standardize input dimensions across the data set, all input images were uniformly resized to 640×640 pixels. The same preprocessing procedure was applied during validation to ensure consistency between training and evaluation phases.

4.3 Evaluation metrics

To assess the performance of the proposed detection and classification model, we adopted several widely used evaluation metrics, including precision, recall, mean Average Precision at IoU threshold 0.5 (mAP@0.5), and mean Average Precision across multiple IoU thresholds (mAP@0.5:0.95). These metrics are formally defined as follows:

$$P = \frac{TP}{TP + FP} \quad (8)$$

Precision measures the accuracy of positive predictions and is calculated as the ratio of true positives (TPs) to the total number of predicted positives, which includes both true positives and false positives (FPs). In other words, precision reflects the proportion of correctly predicted positive

instances out of all instances predicted as positive by the model:

$$R = \frac{TP}{TP + FN} \quad (9)$$

Recall evaluates the model's ability to identify all relevant positive instances. It is defined as the ratio of true positives to the total number of actual positives, which includes both true positives and false negatives (FNs). False negatives refer to instances that are actually positive but were incorrectly classified as negative:

$$mAP = \frac{\sum_{i=1}^n AP_i}{n} \quad (10)$$

Mean Average Precision (mAP) provides an aggregate measure of model performance by combining precision and recall across different classes. Specifically, mAP is computed as the mean of the Average Precision (AP) values for each class. The mAP@0.5 score is calculated using a fixed Intersection over Union (IoU) threshold of 0.5, while mAP@0.5:0.95 represents the average of mAP scores computed at IoU thresholds from 0.5 to 0.95 in 0.05 increments, offering a more comprehensive evaluation of the model's detection accuracy under varying overlap criteria.

5 Experiments

In this section, we provide a systematic analysis of the proposed modules and the refined model. First, we evaluated the performance of the improved model across multiple data sets and conducted detailed comparisons with several state-of-the-art approaches, with particular emphasis on benchmarking against the mainstream YOLO series models across four data sets. Subsequently, we validated the effectiveness of the CL-Concat and C3k2-ARC modules individually and carried out extensive ablation experiments. Moreover, we deployed the model on a Raspberry Pi 5, and the results confirm that, despite the introduced enhancements, the model is still able to maintain comparable inference speed to YOLO11 on lightweight hardware. Finally, we visualized the detection results to more intuitively demonstrate the performance improvements achieved by the refined model.

5.1 Comparison of TOE-YOLO to state-of-the-art methods on the VisDrone2019 data set

A comprehensive comparison between TOE-YOLO and various state-of-the-art object detection methods was conducted using the VisDrone2019 data set. As presented in Table 1 and Fig. 7, the evaluation includes key performance

Table 1 Comparison of TOE–YOLO to state-of-the-art methods on the VisDrone2019 data set

Model	Backbone	GFLOPs	Parameters (M)	mAP50 (%)
SSD [24]	ResNet-50 [43]	62.70	26.30	10.60
DetNet [25]	ResNet-50 [43]	44.60	7.60	29.23
RefineDet [26]	VGG-16 [44]	34.40	11.80	21.37
EfficientDet [27]	EfficientDet-D1 [27]	6.10	6.63	21.20
RetinaNet [28]	ResNet-50-FPN [45]	75.50	21.30	25.50
CenterNet [29]	ResNet-50 [43]	74.90	31.20	29.00
CornerNet [30]	Hourglass-104 [46]	234.00	187.00	34.10
Cascade RCNN [31]	ResNet-50-FPN [45]	236.00	69.29	32.60
TOOD [32]	ResNet-50 [43]	199.00	32.40	33.90
ATSS [33]	ResNet-50-FPN-DyHead [47]	110.00	38.91	33.80
Faster-RCNN [12]	VGG-16 [44]	91.40	41.50	21.90
Double-Head R-CNN [34]	ResNeXt-101 [43]	393.37	47.12	33.40
CSC-YOLO-S [19]	CSPDarkNet [48]	55.10	11.50	33.80
TPH-YOLO [18]	CSPDarkNet [48]	145.70	60.40	36.20
YOLOX-S [35]	CSPDarkNet [48]	26.80	8.90	32.40
YOLOv3-tiny [14]	Darknet-19 [14]	19.10	12.10	23.50
YOLOv5n [36]	CSPDarkNet [48]	7.20	2.51	31.70
YOLOv6n [37]	CSPDarkNet [48]	11.90	4.24	29.70
YOLOv8n [38]	CSPDarkNet [48]	8.90	3.15	32.90
YOLOv9t [39]	CSPDarkNet [48]	8.50	2.12	32.80
YOLOv10n [40]	CSPDarkNet [48]	8.40	2.71	32.00
YOLO12n [41]	CSPDarkNet [48]	6.70	2.60	32.10
YOLO11n [42]	CSPDarkNet [48]	6.60	2.62	32.20
TOE-YOLO	CSPDarkNet [48]	6.60	2.53	33.80

indicators, such as mAP50, mAP50:95, model parameters, and GFLOPs.

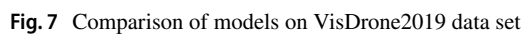
Compared with a range of advanced single-stage and two-stage detectors, including RefineDet, EfficientDet, DetNet, RetinaNet, Faster R-CNN, SSD, Cascade R-CNN, Double-Head R-CNN, and CenterNet. TOE–YOLO demonstrates clear advantages in detection accuracy, model compactness, and computational efficiency. Although TOE–YOLO achieves slightly lower or comparable accuracy relative to models, such as CornerNet, TOOD, and ATSS, these methods require significantly more computational resources, with their parameter counts and GFLOPs exceeding those of TOE–YOLO by several times. On the other hand, when compared with advanced one-stage detectors in the YOLO family, TOE–YOLO demonstrates superior performance in both accuracy and model efficiency. Although its accuracy is 2.4% lower than TPH–YOLO, respectively, the parameter count and GFLOPs of TPH–YOLO are approximately 20 times larger than those of TOE–YOLO. Moreover, in comparison with the smallest models across different YOLO versions, TOE–YOLO achieves higher detection accuracy while maintaining one of the lowest computational footprints in terms of both parameters and GFLOPs.

Therefore, in practical applications, TOE–YOLO is more suitable for aerial image detection tasks compared to

previous YOLO versions and other single-stage object detection methods, and it is well-suited for resource-constrained environments.

5.2 Comparison of TOE–YOLO to state-of-the-art methods on the UAV–ROD data set

A performance evaluation of TOE–YOLO was conducted on the UAV–ROD data set against a range of cutting-edge object detection models. As summarized in Table 2 and Fig. 8, the comparison considers mAP50, GFLOPs, and model parameters. TOE–YOLO delivers an mAP50 score that is 1.3% higher than that of the traditional two-stage detector Faster R-CNN while utilizing over 28 times fewer parameters and computational resources. In addition, Relative to the advanced single-stage detection model YOLOv8s, TOE–YOLO achieves equivalent accuracy with only one-fourth of the GFLOPs. Notably, among all compared methods, such as RetinaNet, TS4-Net, CFC-Net, and other YOLO variants, TOE–YOLO attains the highest detection precision while maintaining an exceptionally lightweight model architecture. This balance between accuracy and efficiency underscores TOE–YOLO's strong potential for deployment in real-time, resource-limited UAV scenarios.



Model	Backbone	GFLOPs	Parameters (M)	mAP50 (%)
RetinaNet [28]	ResNet-50 [43]	36.30	9.20	97.70
Faster R-CNN [12]	ResNet-50 [43]	172.30	34.60	98.00
TS4-Net [49]	ResNet-50 [43]	37.60	9.40	98.10
CFC-Net [50]	ResNet-50 [43]	37.50	9.40	99.30
YOLOv3 [14]	Darknet53 [14]	283.00	103.00	99.30
YOLOv5n [36]	CSPDarkNet [48]	7.20	2.51	99.00
YOLOv5s [36]	CSPDarkNet [48]	24.00	9.12	99.30
YOLOv5m-CSL [36]	CSPDarkNet [48]	20.80	6.10	94.30
YOLOv5m [36]	CSPDarkNet [48]	48.20	20.80	99.30
YOLOv6n [37]	CSPDarkNet [48]	11.90	4.24	99.20
YOLOv6s [37]	CSPDarkNet [48]	42.30	11.80	99.20
YOLOv8n [38]	CSPDarkNet [48]	8.90	3.15	99.20
YOLOv8s [38]	CSPDarkNet [48]	28.80	11.10	99.30
YOLOv9t [39]	CSPDarkNet [48]	8.50	2.12	99.30
YOLOv10n [40]	CSPDarkNet [48]	8.40	2.71	99.10
YOLO12n [41]	CSPDarkNet [48]	6.70	2.60	99.20
YOLO11n [42]	CSPDarkNet [48]	6.60	2.62	99.20
TOE-YOLO	CSPDarkNet [48]	6.60	2.53	99.30

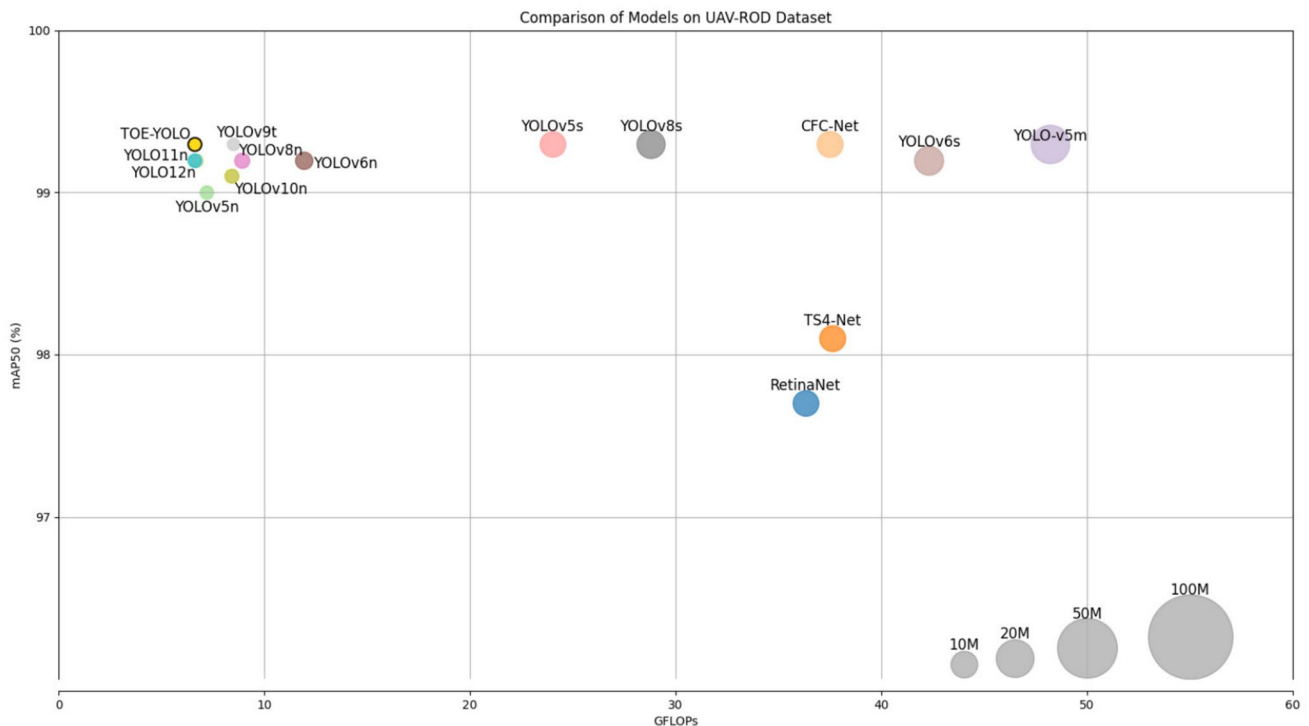


Fig. 8 Comparison of models on UAV-ROD data set

5.3 Comparison of TOE-YOLO to state-of-the-art methods on the CARPK data set

An evaluation of TOE-YOLO was conducted on the CARPK data set, comparing its performance against several state-of-the-art object detection approaches. The results summarized in Table 3 and Fig. 9, highlight TOE-YOLO's effectiveness in terms of mAP50, GFLOPs, and parameter count. Compared to the traditional two-stage detector Faster R-CNN, TOE-YOLO achieves an 11.3% improvement in accuracy while reducing both parameters and GFLOPs by more than 14 times. When benchmarked against advanced methods, such as R-FCN, SSD, FSSD512, GFLV2, VFNet, QueryDet, FCOS, ATSS, and CZ Det, TOE-YOLO attains the highest detection accuracy while maintaining a significantly more compact and computationally efficient model. Furthermore, relative to single-stage detectors, TOE-YOLO surpasses most existing YOLO variants in accuracy, demonstrating its superiority in balancing performance and efficiency.

5.4 Comparison of TOE-YOLO to state-of-the-art methods on the UAVDT data set

An extensive performance comparison of TOE-YOLO was carried out on the UAVDT data set, involving a variety of state-of-the-art object detection frameworks. Table 4 and Fig. 10 summarizes the comparative results based on the

mAP50 metric. The experimental findings demonstrate that TOE-YOLO outperforms the majority of both two-stage and single-stage detectors on this data set. Specifically, when compared to the latest single-stage detectors, such as YOLOv8n, YOLOv10n, and YOLOv12n, TOE-YOLO achieves higher accuracy by margins of 1.6%, 0.7%, and 3.2%, respectively, while maintaining a smaller model size and lower computational complexity in terms of parameters and GFLOPs. In comparison with two-stage detectors, including Double-Head R-CNN, Faster R-CNN, and Cascade R-CNN, TOE-YOLO delivers substantially higher accuracy with significantly reduced parameter counts and GFLOPs. Although the accuracy of TOE-YOLO is slightly below that of QueryDet and CZ Det, it requires far fewer computational resources, and considering that these two-stage methods are less suitable for real-time UAV applications due to their larger model sizes and slower inference speeds, TOE-YOLO offers a more balanced trade-off. Overall, these advantages make TOE-YOLO a compelling choice for real-time object detection tasks in UAV scenarios.

5.5 Comparison with state-of-the-art methods on multiple data sets (mAP_{50:95})

We compared the proposed TOE-YOLO model with the state-of-the-art YOLO framework in terms of mAP_{50:95} across multiple benchmark data sets. As shown in Table 5,

Table 3 Comparison of TOE-YOLO to state-of-the-art methods on the CARPK data set

Model	Backbone	GFLOPs	Parameters (M)	mAP50 (%)
Faster R-CNN [12]	VGG-16 [44]	91.40	41.50	84.80
R-FCN [13]	ResNet-101 [43]	136.20	64.42	86.13
SSD [24]	VGG-16 [44]	47.74	23.88	82.72
FSSD512 [51]	VGG-16 [44]	38.12	34.13	87.59
FCOS [52]	ResNet-101 [43]	85.40	46.00	95.52
ATSS [33]	ResNet-101 [43]	197.09	31.00	95.94
GFLV2 [53]	ResNet-101 [43]	239.32	37.74	94.91
VFNet [54]	VGG-16 [44]	119.21	127.12	94.97
QueryDet [55]	ResNet-50 [43]	125.40	37.74	93.96
CZ Det [56]	ResNet-50 [43]	210.00	45.90	92.18
YOLOv3 [14]	Darknet53 [14]	283.00	103.00	86.01
YOLOv4 [48]	CSPDarkNet [48]	100.60	52.50	86.70
YOLOv5n [36]	CSPDarkNet [48]	7.20	2.51	93.90
YOLOv5s [36]	CSPDarkNet [48]	24.00	9.12	94.20
YOLOv6n [37]	CSPDarkNet [48]	11.90	4.24	94.10
YOLOv8n [38]	CSPDarkNet [48]	8.90	3.15	95.30
YOLOv9t [39]	CSPDarkNet [48]	8.50	2.12	95.50
YOLOv10n [40]	CSPDarkNet [48]	8.40	2.71	95.70
YOLO12n [41]	CSPDarkNet [48]	6.70	2.60	95.00
YOLO11n [42]	CSPDarkNet [48]	6.60	2.62	95.90
TOE-YOLO	CSPDarkNet [48]	6.60	2.53	96.10

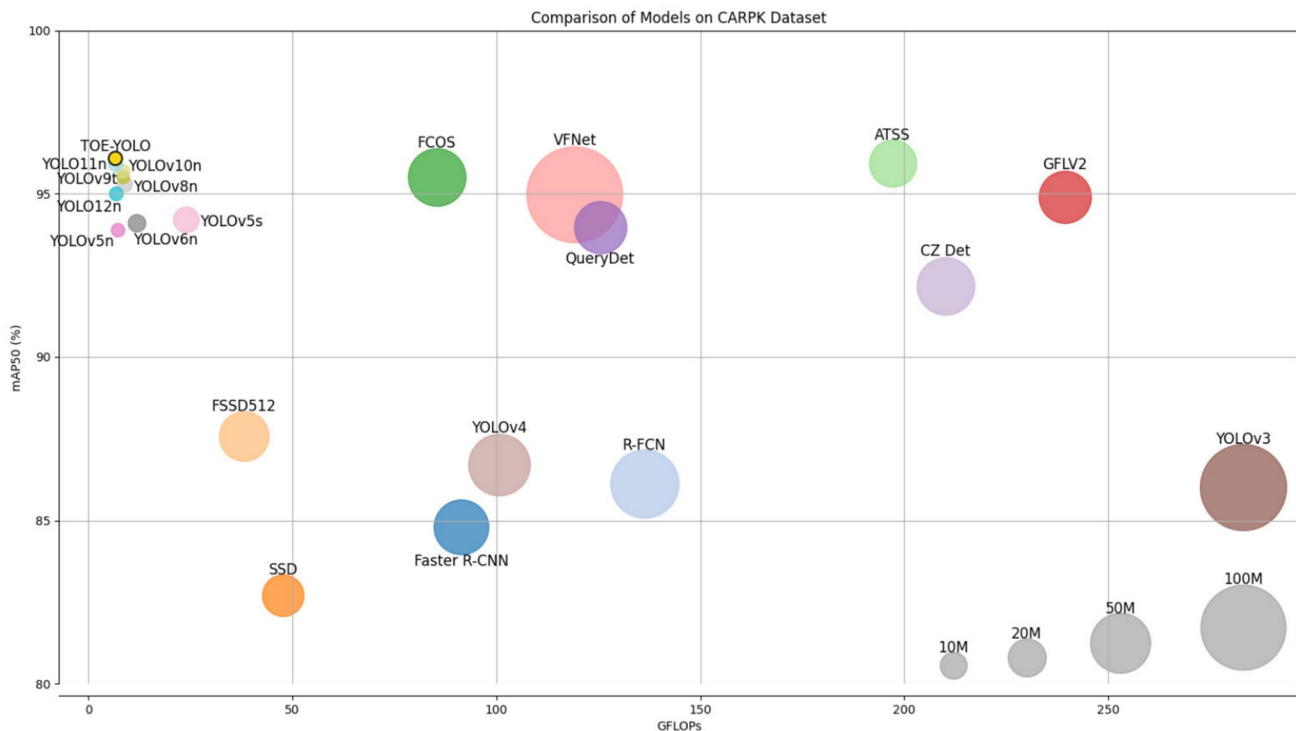
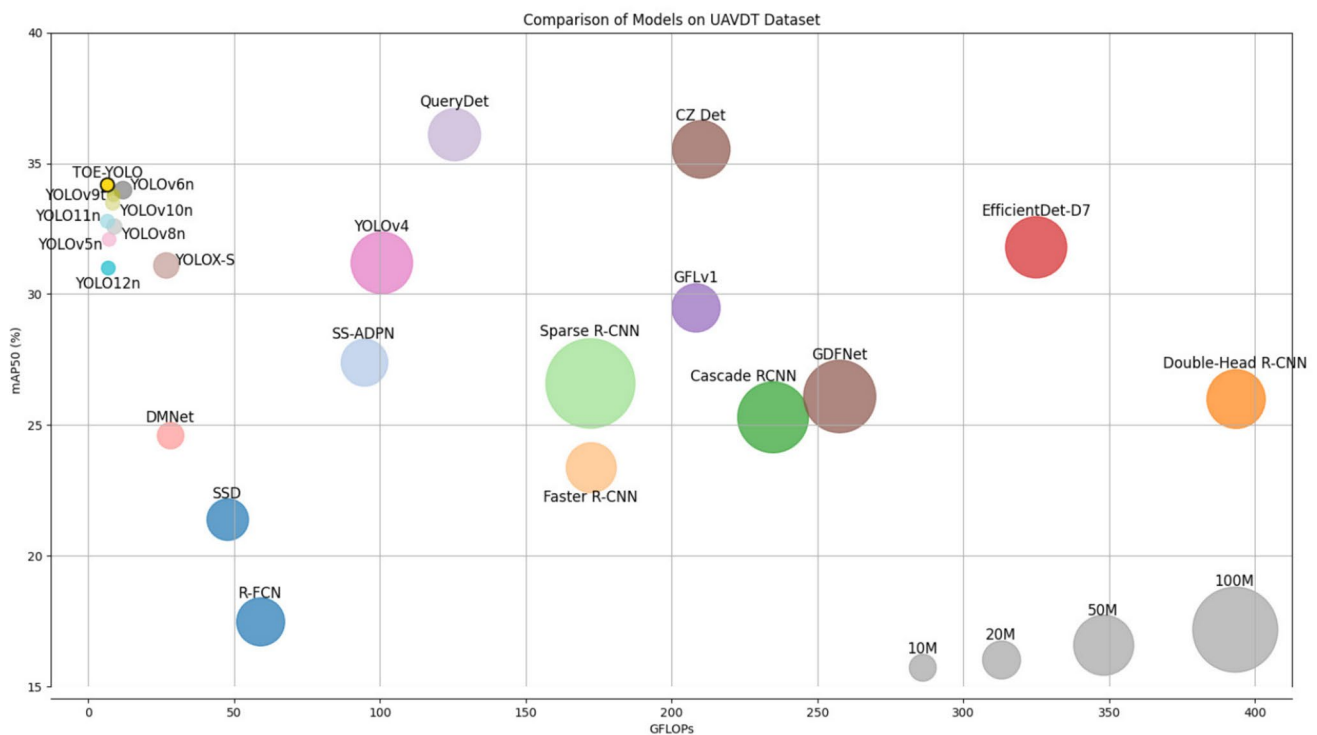
**Fig. 9** Comparison of models on CARPK data set

Table 4 Comparison of TOE–YOLO to state-of-the-art methods on the UAVDT data set

Model	Backbone	GFLOPs	Parameters (M)	mAP50 (%)
RetinaNet [28]	ResNet-50 [43]	36.30	9.20	97.70
Faster-RCNN [12]	ResNet-50 [43]	172.30	34.60	98.00
TS4-Net [49]	ResNet-50 [43]	37.60	9.40	98.10
CFC-Net [50]	ResNet-50 [43]	37.50	9.40	99.30
YOLOv3 [14]	Darknet53 [14]	283.00	103.00	99.30
YOLOv5n [36]	CSPDarkNet [48]	7.20	2.51	99.00
YOLOv5s [36]	CSPDarkNet [48]	24.00	9.12	99.30
YOLO-v5m-CSL [36]	CSPDarkNet [48]	20.80	6.10	94.30
YOLO-v5m [36]	CSPDarkNet [48]	48.20	20.80	99.30
YOLOv6n [37]	CSPDarkNet [48]	11.90	4.24	99.20
YOLOv6s [37]	CSPDarkNet [48]	42.30	11.80	99.20
YOLOv8n [38]	CSPDarkNet [48]	8.90	3.15	99.20
YOLOv8s [38]	CSPDarkNet [48]	28.80	11.10	99.30
YOLOv9t [39]	CSPDarkNet [48]	8.50	2.12	99.30
YOLOv10n [40]	CSPDarkNet [48]	8.40	2.71	99.10
YOLO12n [41]	CSPDarkNet [48]	6.70	2.60	99.20
YOLO11n [42]	CSPDarkNet [48]	6.60	2.62	99.20
TOE-YOLO	CSPDarkNet [48]	6.60	2.53	99.30

**Fig. 10** Comparison of models on UAVDT data set

TOE–YOLO consistently achieves superior performance on all four data sets. Specifically, compared to YOLO11n, TOE–YOLO achieves a 1.3% improvement on Vis-Drone2019, 0.7% on UAVDT, 0.5% on CAPRK, and 0.2%

on UAV–ROD. Moreover, TOE–YOLO substantially outperforms other advanced YOLO models, further demonstrating the effectiveness and superiority of the proposed approach.

Table 5 Comparison with state-of-the-art methods on multiple data sets (mAP_{50:95})

Model	VisDrone2019 mAP _{50:95} (%)	UAVDT	CAPRK	UAV-ROD
YOLOv5n [36]	0.181	0.186	0.656	0.910
YOLOv6n [37]	0.172	0.198	0.660	0.912
YOLOv8n [38]	0.190	0.188	0.648	0.914
YOLOv9t [39]	0.193	0.193	0.649	0.914
YOLOv10n [40]	0.184	0.194	0.660	0.910
YOLO12n [41]	0.186	0.172	0.631	0.913
YOLO11n [42]	0.184	0.194	0.657	0.913
TOE-YOLO	0.197	0.201	0.662	0.915

Table 6 Performance analysis of the CL-Concat module

Model	MAP50	MAP50–95	Gflops	Parameter
Concat	0.322	0.184	6.6	2.60M
Concat + channel attention	0.326	0.189	6.6	2.67M
Concat + local attention	0.328	0.190	6.6	2.71M
CL-Concat	0.330	0.192	6.6	2.71M

Table 7 Performance analysis of the C3k2–ARC module

Model	MAP50	MAP50–95	Gflops	Parameter
C3k2	0.322	0.184	6.6	2.60M
C3k2 + ARConv	0.329	0.191	6.6	2.50M
C3k2 + GhostConv	0.328	0.189	6.5	2.44M
C3k2-ARC	0.330	0.192	6.5	2.48M

5.6 Performance analysis of the CL-Concat module and C3k2–ARC module

To further evaluate the effectiveness of the different components in the CL-Concat and C3k2–ARC modules, we conducted a detailed module analysis and experimental validation.

First, for the CL-Concat module, we compared the performance of using only Concat, adding Channel Attention, adding Local Attention, and the complete CL-Concat As shown in Table 6. The experimental results show that introducing CL-Concat achieves the highest mAP, while the GFLOPs

remain unchanged. This validates the effectiveness of our proposed CL-Concat module in enhancing feature fusion quality without increasing computational cost.

Second, for the C3k2–ARC module, we compared using only C3k2, adding ARConv, adding GhostConv, and the complete C3k2–ARC As shown in Table 7. The results indicate that the full C3k2–ARC module achieves the highest accuracy, while both GFLOPs and the number of parameters are lower than some alternative configurations. This further demonstrates the effectiveness of the C3k2–ARC module in enhancing feature extraction capability and contributing to a lightweight model design.

5.7 Ablation studies

The ablation study presented in Table 8 is conducted on the VisDrone2019 data set to evaluate the individual and combined contributions of the proposed C3k2–ARC and CL-Concat modules to the performance of TOE–YOLO. Starting from the baseline model without any additional components, introducing only the C3k2–ARC module improves the precision to 0.445 and recall to 0.329, while mAP@0.5 and mAP@0.5:0.95 increase by 0.8% compared to the baseline. Similarly, incorporating only the CL-Concat module yields a precision of 0.443 and a recall of 0.328, with both mAP@0.5 and mAP@0.5:0.95 also increasing by 0.8%. When both modules are integrated into the complete TOE–YOLO architecture, the model achieves the best performance across all evaluation metrics: precision increases to 0.450, recall to 0.337, mAP@0.5 reaches 33.8%, and mAP@0.5:0.95 rises to 19.7%. Compared with the baseline, this represents an improvement of 0.021 in precision, 0.014 in recall, and increases of 1.6 and 1.3 percentage points in mAP@0.5 and mAP@0.5:0.95, respectively. These results demonstrate that both C3k2–ARC and CL-Concat have positive effects on performance and exhibit strong complementarity when used together.

5.8 Performance evaluation on Raspberry Pi 5

To verify that our model can maintain performance comparable to YOLO11 on resource-constrained hardware, we conducted inference tests on a Raspberry Pi. Using 500 images from the VisDrone data set, we resized the input images to 640×640 , 320×320 , and 160×160 , and compared the FPS and inference times. The results, shown in Table 9, indicate

Table 8 Ablation study of TOE–YOLO on the VisDrone2019 data set

Model	+C3k2-ARC	+CL-Concat	Precision	Recall	mAP50 (%)	mAP50–95 (%)
Baseline	–	–	0.429	0.323	0.322	0.184
	✓	–	0.445	0.329	0.330	0.192
	–	✓	0.443	0.328	0.330	0.192
TOE-YOLO	✓	✓	0.450	0.337	0.338	0.197

Table 9 Performance Evaluation on Raspberry Pi 5 between YOLO11n and TOE-YOLO

	640 × 640		320 × 320		160 × 160	
	FPS	Reference time (s)	FPS	Reference time (s)	FPS	Reference time (s)
Yolo11n	2.44	0.429	5.98	0.167	12.90	0.077
TOE-YOLO	2.91	0.344	6.41	0.156	15.50	0.065

that our model achieves faster inference and higher FPS than YOLO11 across all input sizes, demonstrating its efficiency and superior performance on lightweight hardware.

5.9 Visualization results

To further validate the generalization capability of TOE-YOLO across different data sets, we conducted evaluations on four data sets: VisDrone2019, UAV-ROD, CARPK, and UAVDT. As illustrated in Figs. 11, 12, 13 and 14, TOE-YOLO consistently demonstrates outstanding detection performance across these data sets, effectively identifying multi-scale rotated objects, especially excelling in small object detection tasks that are common in UAV imagery. Compared with the baseline model, TOE-YOLO exhibits a significant improvement in overall detection accuracy.

Figure 11 presents the visualization results on the VisDrone2019 data set. As shown in subfigures a–d, TOE-YOLO achieves higher detection rates for small objects, such as cars and motorcycles, which often appear in various orientations. Figure 12 displays the results on the

UAV-ROD data set, where TOE-YOLO achieves a lower false positive rate in detecting vehicles located in darker edge regions of the images while maintaining high recall in densely populated scenes. Figure 13 illustrates the detection performance on the CARPK data set, where TOE-YOLO successfully detects partially occluded targets and exhibits a lower duplication rate in identifying densely arranged small vehicles. Finally, Fig. 14 shows the results on the UAVDT data set, where TOE-YOLO demonstrates higher accuracy in detecting extremely small and distant targets.

These results highlight the superior performance of TOE-YOLO in object detection tasks, particularly in handling multi-scale rotated targets and small objects under complex scenarios. This can be largely attributed to the enhancement of the traditional C3k2 structure through the proposed C3k2-ARC module, which incorporates the ARCBLOCK consisting of Adaptive Rotated Convolution (ARConv) and GhostConv. This design strengthens the model's sensitivity to targets with arbitrary orientations while maintaining computational efficiency, thereby improving rotation robustness and feature representation. In addition, the CL-Concat module, by integrating both channel attention

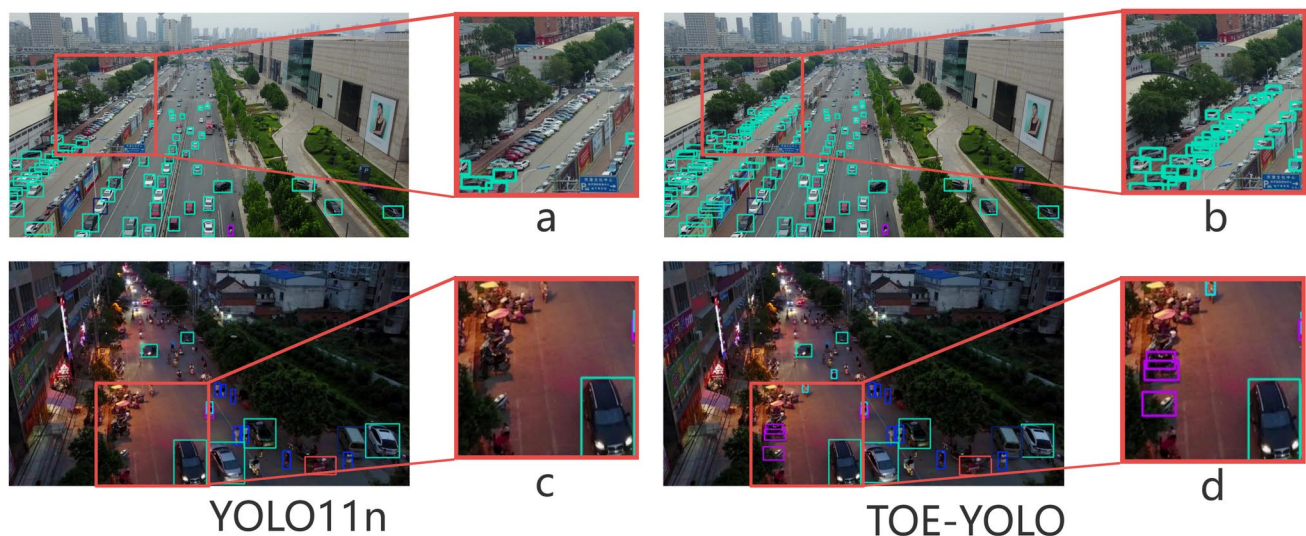


Fig. 11 a–d Visualization of detection results on the VisDrone2019 data set using the pre-trained TOE-YOLO model. Subfigures (a, c) present the predictions of the baseline YOLO11n, while (b, d) display the corresponding results from TOE-YOLO. Predicted bounding

boxes for *cars* and *bikes* are highlighted in green and purple, respectively. Zoomed-in insets are included to emphasize finer detection differences between the two models

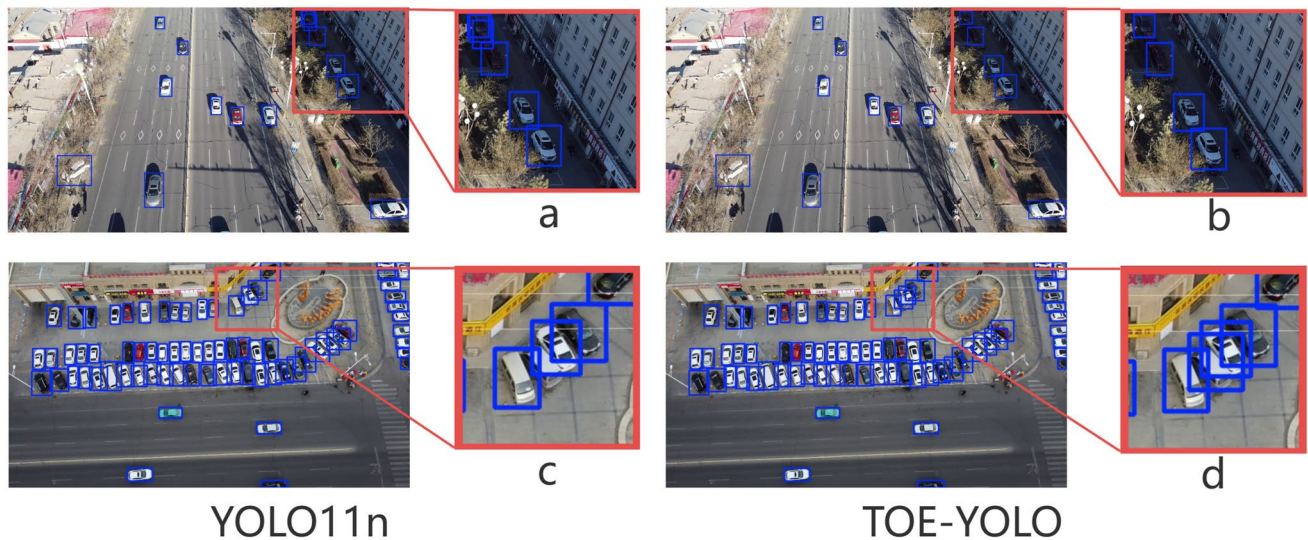


Fig. 12 a–d Detection visualizations on the UAV-ROD data set using the pre-trained TOE-YOLO model. Subfigures (a, c) illustrate the predictions made by YOLO11n, while (b, d) display the corre-

sponding outputs from TOE-YOLO. Blue bounding boxes denote the detected cars. Zoomed-in sections are included to highlight differences in detection performance

Fig. 13 a–d Visual comparison of detection results on the CAPRK data set between YOLO11n and TOE-YOLO. Figures (a, c) show outputs from YOLO11n, whereas (b, d) present those from TOE-YOLO. Predicted cars are marked with blue bounding boxes. Enlarged views emphasize the improved localization accuracy of TOE-YOLO

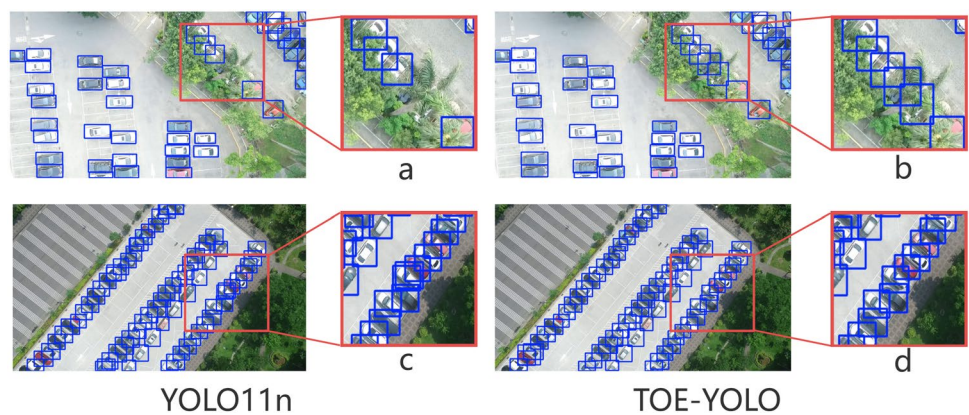
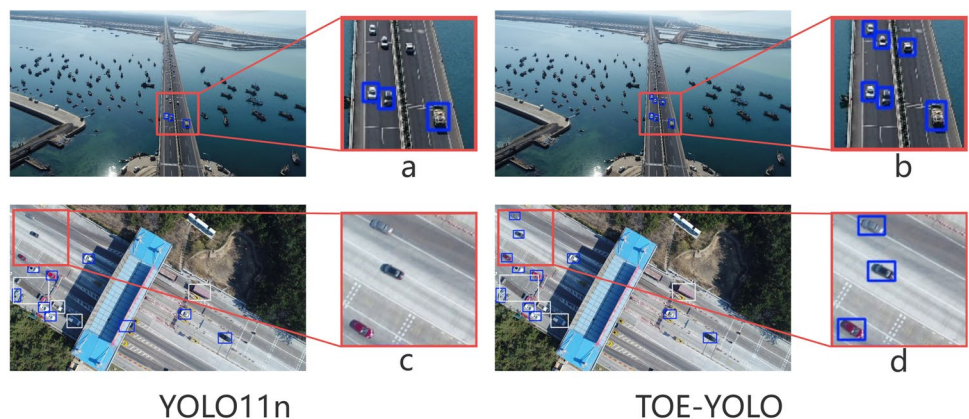


Fig. 14 a–d Qualitative results on the UAV-DT data set comparing YOLO11n and TOE-YOLO. Detection outputs from YOLO11n are shown in (a, c), while those from TOE-YOLO appear in (b, d). All predicted cars are enclosed in blue bounding boxes. Close-up views are provided to showcase the refinement in detection capability



and location attention mechanisms, effectively mitigates semantic inconsistencies and information redundancy during multi-scale feature fusion. This enhances the recognizability

and spatial awareness of small objects. The synergy of these modules significantly improves TOE-YOLO's

generalization capability and detection accuracy in complex UAV image scenarios.

6 Conclusions

This paper proposes a novel object detection model named TOE-YOLO, which is specifically designed to improve the efficiency and accuracy of multi-scale object detection in UAV imagery. By incorporating a rotation-aware feature extraction module and an enhanced feature fusion strategy, TOE-YOLO achieves notable performance improvements in detecting multi-scale targets while slightly reducing the overall model size. This ensures a balanced trade-off between accuracy and computational efficiency.

To address the accuracy degradation caused by the rotational properties of objects in aerial imagery, this work introduces the improved C3k2-ARC module, which enhances the model's capacity to extract and represent rotation-sensitive features through angle-aware operations and direction-adaptive encoding. In addition, to alleviate the semantic inconsistency and spatial detail loss often encountered during conventional feature fusion, we propose the CL-Concat module. This module employs a two-stage attention mechanism that first performs channel-level refinement, followed by spatial-level enhancement. By combining channel and location attention in a complementary and sequential manner, CL-Concat effectively preserves essential spatial cues while improving the semantic alignment of multi-scale features.

Extensive experiments on four benchmark data sets, namely, VisDrone2019, UAV-ROD, CARPK and UAVDT, demonstrate that TOE-YOLO consistently outperforms various baseline models in terms of detection accuracy. Notably, these improvements are achieved with a slight reduction in model parameters and computational cost. The results confirm the effectiveness and robustness of TOE-YOLO in addressing the challenges of detecting multi-scale and rotated objects in UAV imagery. This makes TOE-YOLO particularly well-suited for real-time aerial applications, where lightweight architecture and high precision are essential for practical deployment.

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Author Contributions Haimin Yan conducted the experiments, performed data analysis, and drafted the initial manuscript. Haimin Yan and Xiangbo Kong designed the experimental methodology. Xiangbo Kong acquired the funding. Haimin Yan, Xiangbo Kong, Tomoyasu Shimada, and Hiroyuki Tomiyama analyzed the conclusions and critically revised the manuscript.

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Data Availability No data sets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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